

# Mathematical Notation Reference

Measure Theory, Set Theory, Probability & Statistics

Personal Reference

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## 1 Introduction

This document serves as a reference for mathematical notations used in Measure Theory, Set Theory, Probability, and Statistics.

## 2 Set Theory

| Notation                  | Explanation                                                 | Purpose                                                                      |
|---------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|
| $x \in A$                 | Element $x$ belongs to set $A$ .                            | Basic set membership.                                                        |
| $A \subseteq B$           | Set $A$ is a subset of set $B$ .                            | Indicates inclusion.                                                         |
| $A \cap B$                | Intersection of sets $A$ and $B$ (elements common to both). | Logical AND operation on sets.                                               |
| $A \cup B$                | Union of sets $A$ and $B$ (elements in either $A$ or $B$ ). | Logical OR operation on sets.                                                |
| $\emptyset$               | The empty set containing no elements.                       | Identity element for union.                                                  |
| $2^A$ or $\mathcal{P}(A)$ | Power set of $A$ (set of all subsets of $A$ ).              | Defining sigma-algebras.                                                     |
| $\forall$                 | Universal Quantifier (For all)                              | Defining properties that must hold for every element (e.g., continuity).     |
| $\exists$                 | Existential Quantifier (There exists)                       | Asserting the existence of a specific object (e.g., existence of a limit).   |
| $f^{-1}(B)$               | Inverse Image                                               | The core tool for defining Random Variables (Mapping values back to events). |

## 3 Measure Theory

| Notation   | Explanation      | Purpose                                                      |
|------------|------------------|--------------------------------------------------------------|
| $\mu, \nu$ | General Measures | Generalizing the concept of “size” (volume, mass, or count). |

| Notation                  | Explanation              | Purpose                                                                            |
|---------------------------|--------------------------|------------------------------------------------------------------------------------|
| $\lambda$                 | Lebesgue Measure         | The standard way to measure length/area/volume on the real line $\mathbb{R}$ .     |
| $\mathcal{B}(\mathbb{R})$ | Borel $\sigma$ -algebra  | The standard collection of “well-behaved” sets on $\mathbb{R}$ used in statistics. |
| $\nu \ll \mu$             | Absolute Continuity      | A condition ensuring that if $\mu$ sees nothing (zero), $\nu$ also sees nothing.   |
| $\frac{d\mu}{dP}$         | Radon-Nikodym Derivative | Formally defining the Probability Density Function (PDF) as a ratio of measures.   |
| $\int_A f d\mu$           | Lebesgue Integral        | Calculating the “average” or “total” over a set $A$ using measure $\mu$ .          |

## 4 Probability & Statistics

| Notation                   | Explanation                 | Purpose                                                                            |
|----------------------------|-----------------------------|------------------------------------------------------------------------------------|
| $\Omega$                   | Sample Space                | The set of all possible “raw” outcomes in an experiment.                           |
| $\mathcal{F}$              | $\sigma$ -algebra           | The collection of “measurable” events or the “information structure” available.    |
| $P$                        | Probability Measure         | A function that assigns a value in $[0, 1]$ to every event in $\mathcal{F}$ .      |
| $\mathbb{I}_A$             | Indicator Function          | Converting a set $A$ into a function (1 if inside, 0 if outside) for calculations. |
| $X_n \xrightarrow{a.s.} X$ | Almost Sure Convergence     | Proving the Strong Law of Large Numbers (the most rigorous convergence).           |
| $X_n \xrightarrow{P} X$    | Convergence in Probability  | The standard convergence for consistent estimators in large samples.               |
| $X_n \xrightarrow{d} X$    | Convergence in Distribution | The foundation of the Central Limit Theorem (limit behavior of distributions).     |
| $\theta$                   | Parameter                   | The “hidden truth” or the target value in a statistical model.                     |

| Notation                    | Explanation             | Purpose                                                                                  |
|-----------------------------|-------------------------|------------------------------------------------------------------------------------------|
| $\hat{\theta}_n$            | Estimator               | The rule or formula used to guess $\theta$ based on $n$ data points.                     |
| $\mathbb{E}[X \mathcal{G}]$ | Conditional Expectation | Predicting $X$ given a specific sub-information (sub- $\sigma$ -algebra) $\mathcal{G}$ . |